Data Structures and Algorithms

LECTURE 04: ALGORITHM ANALYSIS

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Alliance with FPT Education

Contents

- Algorithmic Complexity
- Time Complexity
- Asymptotic notations
- Brute Force

Algorithm Analysis

- Why should we analyze algorithms?
  - Predict the **resources** the algorithm will need
    - Computational time (**CPU** consumption)
    - Memory space (**RAM** consumption)
    - Communication **bandwidth** consumption
    - **Hard disk** operations

Problem: Get Number of Steps

- Calculate maximum steps to find the result

```
long getOperationsCount(int n) {
  long counter = 0;
  for (int i = 0; i < n; i++)
      for (int j = 0; j < n; j++)
      counter++;
  return counter;
}
```

Solution:
T(n) = 3(n ^ 2) + 3n + 3

- The input(**n**) of the function is the main source of steps growth

Simplifying Step Count

- Some parts of the equation **grow much faster** than others
  - $T(n) = \mathbf{3(n^2)} + 3n + 3$
  - We can **ignore** some part of this equation
  - Higher terms **dominate** lower terms – $\mathbf{n > 2}$, $\mathbf{n^2 > n}$, $\mathbf{n^3 > n^2}$
  - Multiplicative constants can be **omitted** – $\mathbf{12n \rightarrow n}$, $\mathbf{2n^2 \rightarrow n^2}$
- The previous solution becomes $\approx \mathbf{n^2}$

Time Complexity

- Worst-case
  - An **upper** bound on the running time
- Average-case
  - **Average** running time
- Best-case
  - The **lower** bound on the running time (the optimal case)

Time Complexity

- Therefore, we need to measure **all** the possibilities:

Time Complexity

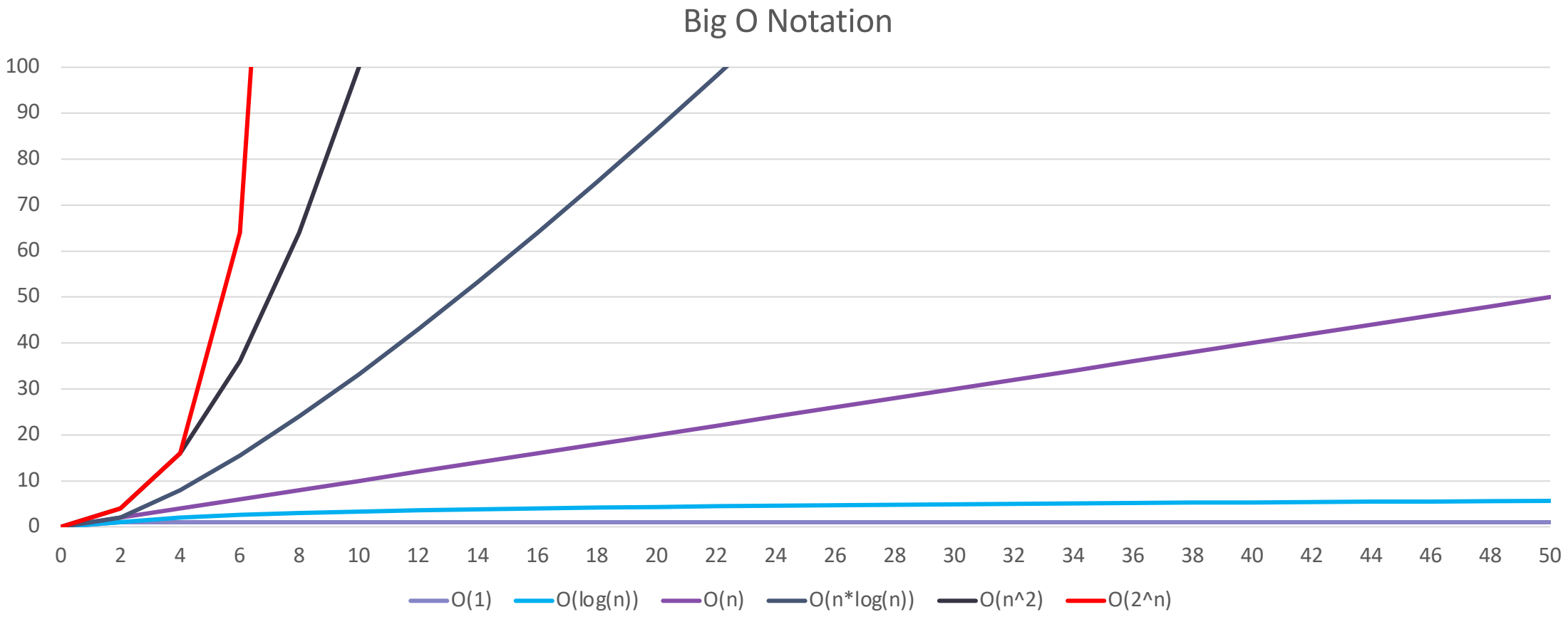
- From the previous chart we can deduce:
  - For smaller size of the input (**n**) we **don't care much for the runtime**. So we measure the time as **n** approaches **infinity**
  - If an algorithm **has to scale**, it **should compute** the result within a **finite and practical time**
  - We're concerned about the **order of an algorithm's complexity**, not the actual time in terms of **milliseconds**

Asymptotic notations

- **Asymptotic notations** are descriptions that allow us to examine an algorithm's running time by expressing its **performance** as the input size, **n**, of an algorithm or a function **f increases**. There are **three** common asymptotic notations:
  - Big **O – O(f(n))**
  - Big **Theta – Θ(f(n))**
  - Big **Omega – Ω(f(n))**

Asymptotic Functions

- Below are some examples of **common algorithmic** grow:



Typical Complexities

Complexity	Notation	Description
constant	O(1)	n = 1 000 → 1-2 operations
logarithmic	O(log n)	n = 1 000 → 10 operations
linear	O(n)	n = 1 000 → 1 000 operations
linearithmic	O(n*log n)	n = 1 000 → 10 000 operations
quadratic	$O(n^2)$	n = 1 000 → 1 000 000 operations
cubic	$O(n^3)$	n = 1 000 → 1 000 000 000 operations
exponential	$O(n^n)$	n = 10 → 10 000 000 000 operations

Time Complexity and Program Speed

- **Demo**

Summary

- Algorithmic **Complexity**
- Time Complexity